

Social Media Fake Review Detection

Presented by :

Farhana Najnin: 20210104109

Zarin Tasnim: 20210104111

Afrain Akhter: 20210104113

Maliha Akter: 20210104122



Table of Contents

- Introduction
- Motivation
- Problem Statement
- Research Objectives
- Model Used
- Dataset
- Data Preprocessing
- Model Training Process
- Evaluation Metrics
- Result Analysis
- Observation & Insights
- Possible Improvements
- Challenges and Limitation
- Conclusion



INTRODUCTION

- Fake reviews on social media mislead consumers and impact businesses.
- Manual detection is time-consuming and ineffective.
- AI-powered systems can automatically detect and filter fake reviews.
- Transformers and Large Language Models (LLMs) have revolutionized text classification tasks.
- Our study compares transformer-based architectures and LLMs for detecting fake reviews.



PROBLEM STATEMENT

- Fake reviews on social media are increasing
- They mislead people and harm businesses.
- Traditional methods can't detect them effectively.
- Fake reviews are written in different languages and styles.
- AI-based models can help detect them automatically.



MOTIVATION

- **Growing Influence of Social Media:** Reviews shape public perception and consumer decisions.
- **Rising Fake Reviews:** Manipulated opinions affect businesses and mislead users.
- **Challenges of Manual Detection:** Fake reviews are often indistinguishable from real ones and spread rapidly.
- **Need for Advanced AI Solutions:** Transformers & LLMs can analyze large-scale review data efficiently.
- **Towards a Trustworthy Digital Space:** Automating fake review detection enhances user trust and platform integrity.



RESEARCH OBJECTIVES

- **Detect Fake Reviews on Social Media** Build a system that accurately identifies fake reviews from social media platforms.
- **Compare Performance of Transformer Models & LLMs** Evaluate and compare the performance of various transformer-based models like mBERT, XLM-R, Bangla-T5, and LLMs for fake review detection.
- **Analyze Effectiveness for Bangla Language** Investigate which model performs best for detecting fake reviews in the Bangla language.
- **Enhance Accuracy with Fine-Tuned Models** Improve the accuracy of fake review detection by fine-tuning pre-trained transformer models and large language models.



MODEL USED

mBERT (Multilingual BERT): Handles multilingual fake review detection.

XLNet (Cross-lingual Model): Works well for cross-language detection.

BanglaBERT: Fine-tuned for Bangla text processing.

Bangla-T5: Specialized for Bangla text generation and classification.

LLM (Large Language Model): Deep contextual understanding.

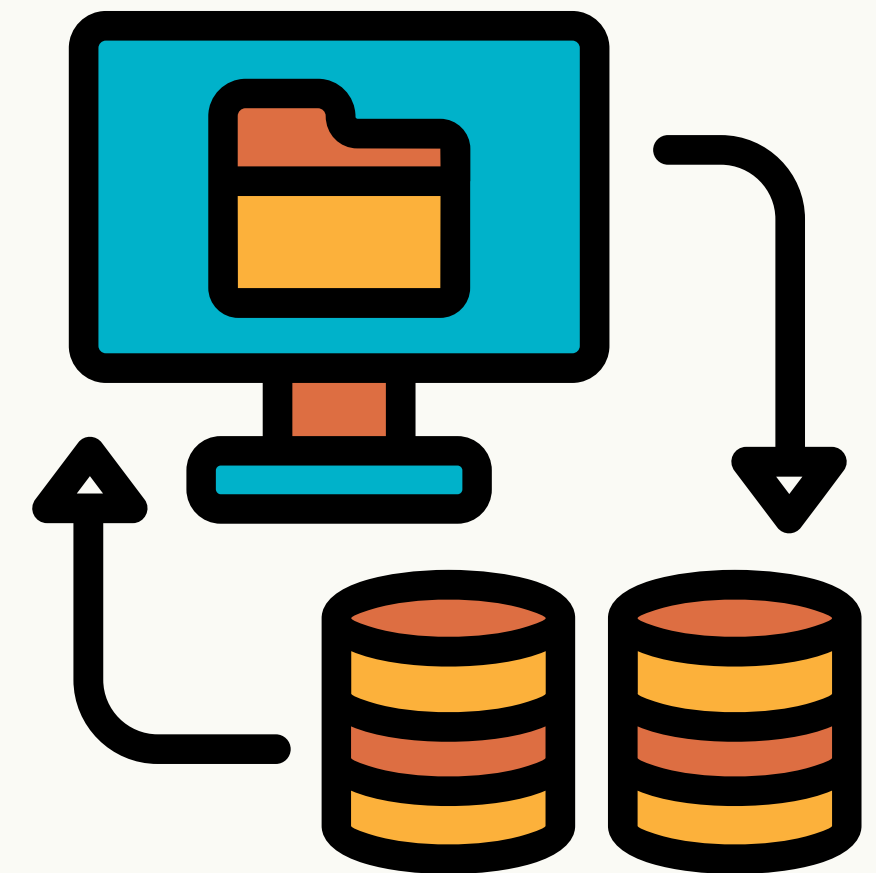
DATASET

- Bengali Fake Review Dataset from Hugging Face.
- Contains real and fake reviews in Bangla.
- Preprocessed by tokenization and text normalization.
- Split into training (80%) and testing (20%).



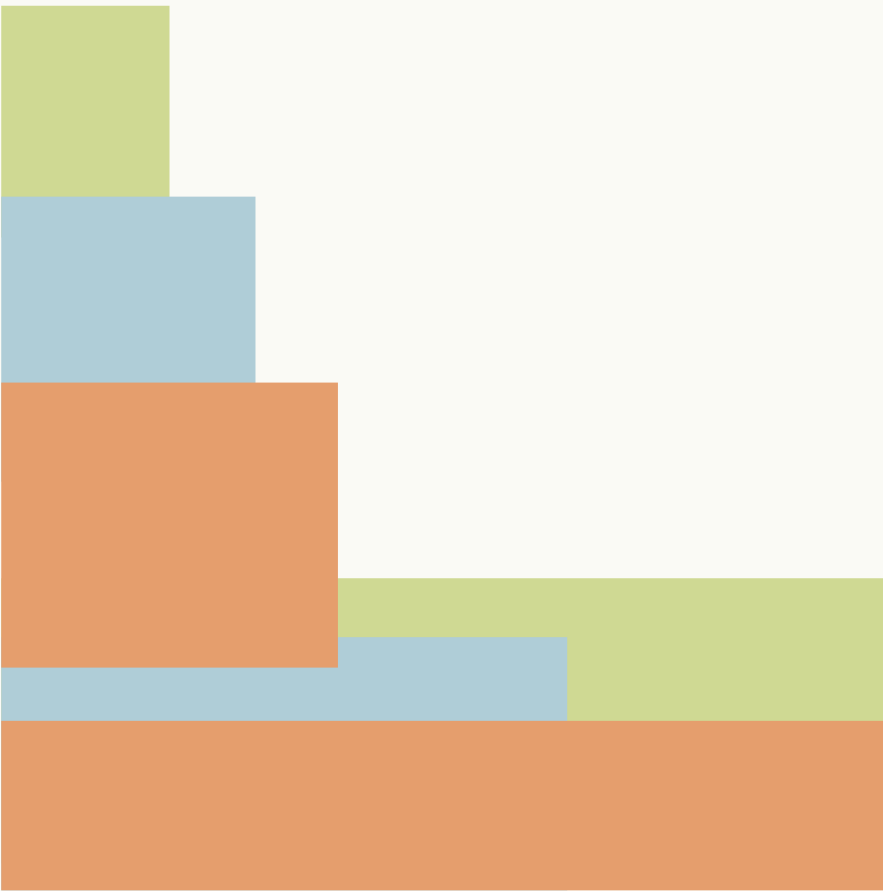
DATA PREPROCESSING

- **Text Cleaning:** Remove special characters, stopwords, and irrelevant symbols.
- **Tokenization:** Use model-specific tokenizers (e.g., mBERT, XLM-R) for Bangla text.
- **Encoding:** Convert tokenized text into numerical representations.
- **Balanced Class Distribution:** Ensure equal distribution of real and fake reviews to prevent bias.
- **Handling Imbalanced Data:** Apply techniques like oversampling or undersampling if needed.
- **Text Normalization:** Convert text to lowercase, remove extra spaces, and handle diacritics.
- **Outlier Removal:** Identify and remove reviews that are too short or not informative.
- **Padding:** Standardize text length for consistent input to models.





MODEL TRAINING PROCESS

- Fine-tuned each model using LoRA (Low-Rank Adaptation).
 - Used **Hugging Face Transformers** and **PEFT (Parameter-Efficient Fine-Tuning)**.
 - Trained models for **3 epochs** with batch size = 4.
 - Optimized using **AdamW optimizer** and **Cross-Entropy Loss**.
- 

EVALUATION METRICS

- **Accuracy:** How often the model correctly classifies reviews.
- **Precision:** Ability to correctly identify fake reviews.
- **Recall:** How well the model captures all fake reviews.
- **F1-Score:** Balance between precision and recall.



RESULT & ANALYSIS

Model	Accuracy	Precision	Recall	F1-Score
XLM-R	94%	94.21%	99.08%	96.58%
mBERT	87.26%	87.68%	86.69%	87.18%
BanglaBERT	97.90%	98.18%	99.34%	98.76%
LLM				
BanglaT5				

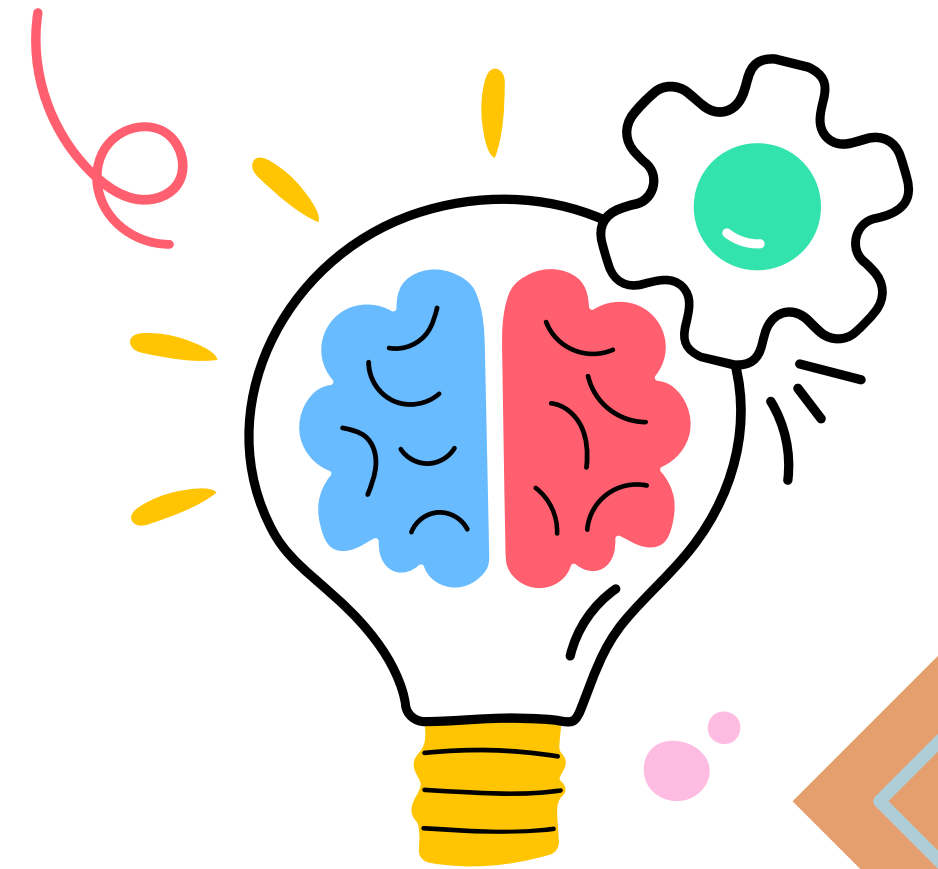


OBSERVATION & INSIGHTS

- **Best performing model:** [BanglaBERT]
- **mBERT & XLM-R:** Worked well for multilingual fake detection.
- **Bangla-T5 & BanglaBERT:** Better suited for Bangla-specific detection.
- **LLM:** Captured deep contextual meaning but required high computational power.
- Trade-off between accuracy and efficiency.

POSSIBLE IMPROVEMENTS

- **LOWER LOSS:** TUNE LEARNING RATE, INCREASE TRAINING DATA.
- **REDUCE OVERFITTING:** USE DROPOUT, WEIGHT DECAY, MORE DATA AUGMENTATION.
- **OPTIMIZE COMPUTATIONAL EFFICIENCY:** USE QUANTIZATION, PRUNING, OR EDGE AI TECHNIQUES.
- **ENHANCE DATASET:** COLLECT MORE DIVERSE REAL-WORLD FAKE REVIEWS.



CHALLENGES & LIMITATIONS

- **OOM Errors on Larger Models:** Attempting to fine-tune LLMs such as Gemma-2B, Llama-2-7B-Chat, and Falcon-7B resulted in CUDA Out-of-Memory errors, even after optimizing batch sizes and clearing GPU cache. Computational Bottlenecks: Even with batch size reduction, memory fragmentation and tensor allocation failures persisted, limiting the feasibility of fine-tuning these models on a 20GB VRAM GPU.
- **Training Time Constraints:** The models that were successfully fine-tuned required significant

Conclusion

- **Large Language Models (LLMs)** enhance fake review detection with deeper contextual understanding.
- **mBERT & XLM-R** are strong multilingual transformers for cross-lingual fake review detection.
- **BanglaBERT & Bangla-T5** perform well for Bangla-specific fake review classification.
- LLMs fine-tuned on domain-specific data improve accuracy and adaptability.
- **Future Work:** Enhance real-time detection, reduce bias, and integrate with edge AI for faster deployment.

